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Lab 3 : Build an Interactive Grade Calculator

Task:

Create

a Lab03 folder with index.html, style.css and app.js. Link app.js using defer

- Add inputs for Student Name and Marks (0–100), plus a "Calculate Grade" button and an empty result <div>.
- On button click, read both values with .value and convert marks using Number().
- Use if / else if / else to assign a grade: A (85+), B (70+), C (50+), F (below 50).
- Show a message in the result div, e.g. "Ali scored 78 → Grade B", using a template literal.
- Colour the result green for a pass and red for a fail.
Bonus: show an error if marks are empty or out of range.

Code Screenshot:

Index.Html:

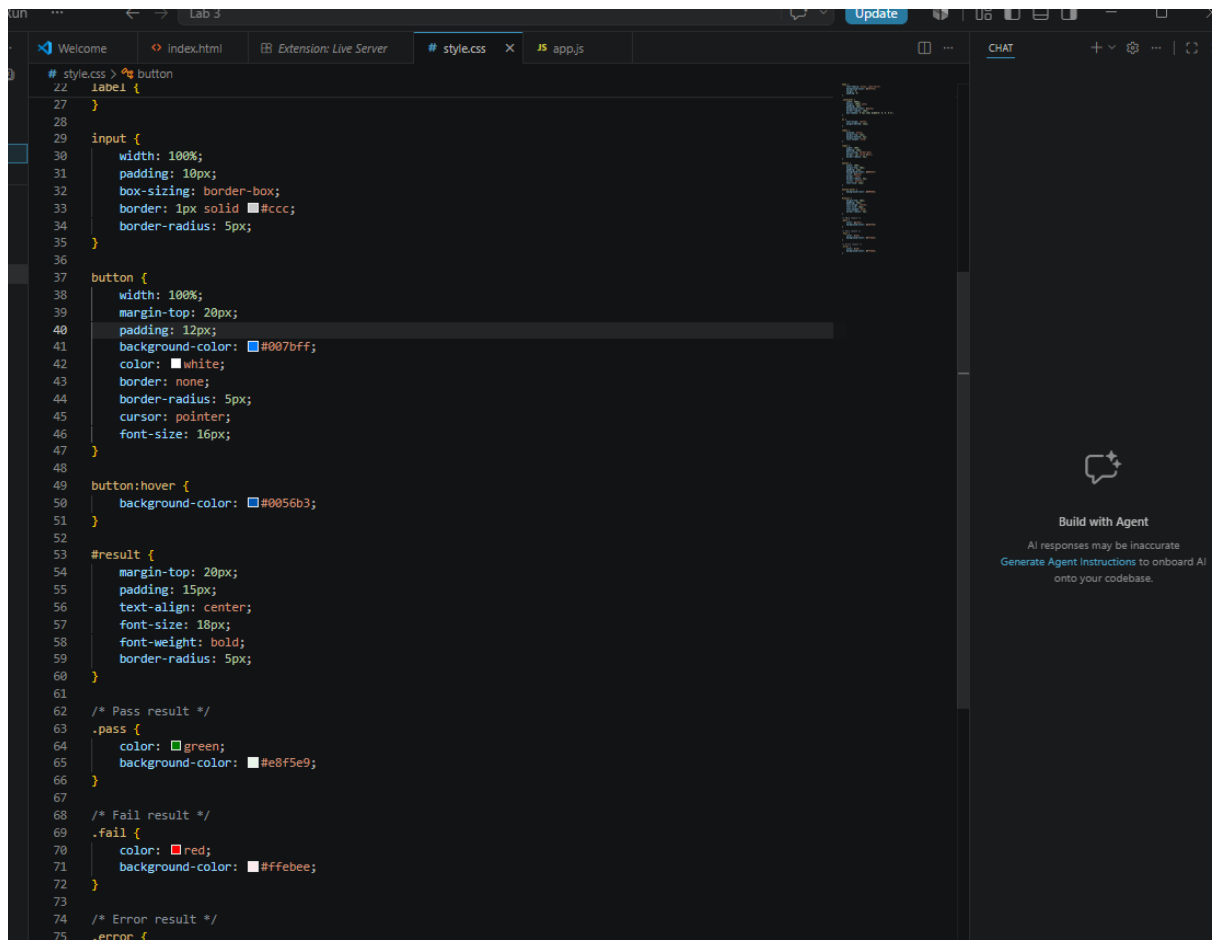
```
...  ← →  Lab 3  Update  [Icons]
Welcome  index.html ×  Extension: Live Server  # style.css  JS app.js  CHAT  + v ⚙ ... | ⌂
index.html > html > body
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6
7   <title>Grade Calculator</title>
8
9   <link rel="stylesheet" href="style.css">
10
11   <!-- JavaScript file linked using defer -->
12   <script src="app.js" defer></script>
13 </head>
14
15 <body>
16
17   <div class="container">
18
19     <h1>Interactive Grade Calculator</h1>
20
21     <label for="studentName">Student Name</label>
22     <input type="text" id="studentName" placeholder="Enter student name">
23
24     <label for="marks">Marks</label>
25     <input
26       type="number"
27       id="marks"
28       placeholder="Enter marks (0-100)"
29       min="0"
30       max="100"
31     >
32
33     <button id="calculateBtn">Calculate Grade</button>
34
35     <div id="result"></div>
36
37   </div>
38
39 </body>
40 </html>
41
```



Build with Agent

AI responses may be inaccurate
Generate Agent Instructions to onboard AI
onto your codebase.

Style.Css:



App.js:

```
Run ... Lab 3 Update
Welcome index.html Extension: Live Server style.css JS app.js x
app.js calculateBtn.addEventListener("click", function () {
  // Check if marks are empty
  if (marksValue === "") {
    resultDiv.textContent = "Please enter marks.";
    resultDiv.className = "error";
    return;
  }

  // Convert marks to Number
  const marks = Number(marksValue);

  // Check if marks are out of range
  if (marks < 0 || marks > 100) {
    resultDiv.textContent = "Marks must be between 0 and 100.";
    resultDiv.className = "error";
    return;
  }

  // Check if student name is empty
  if (studentName.trim() === "") {
    resultDiv.textContent = "Please enter student name.";
    resultDiv.className = "error";
    return;
  }

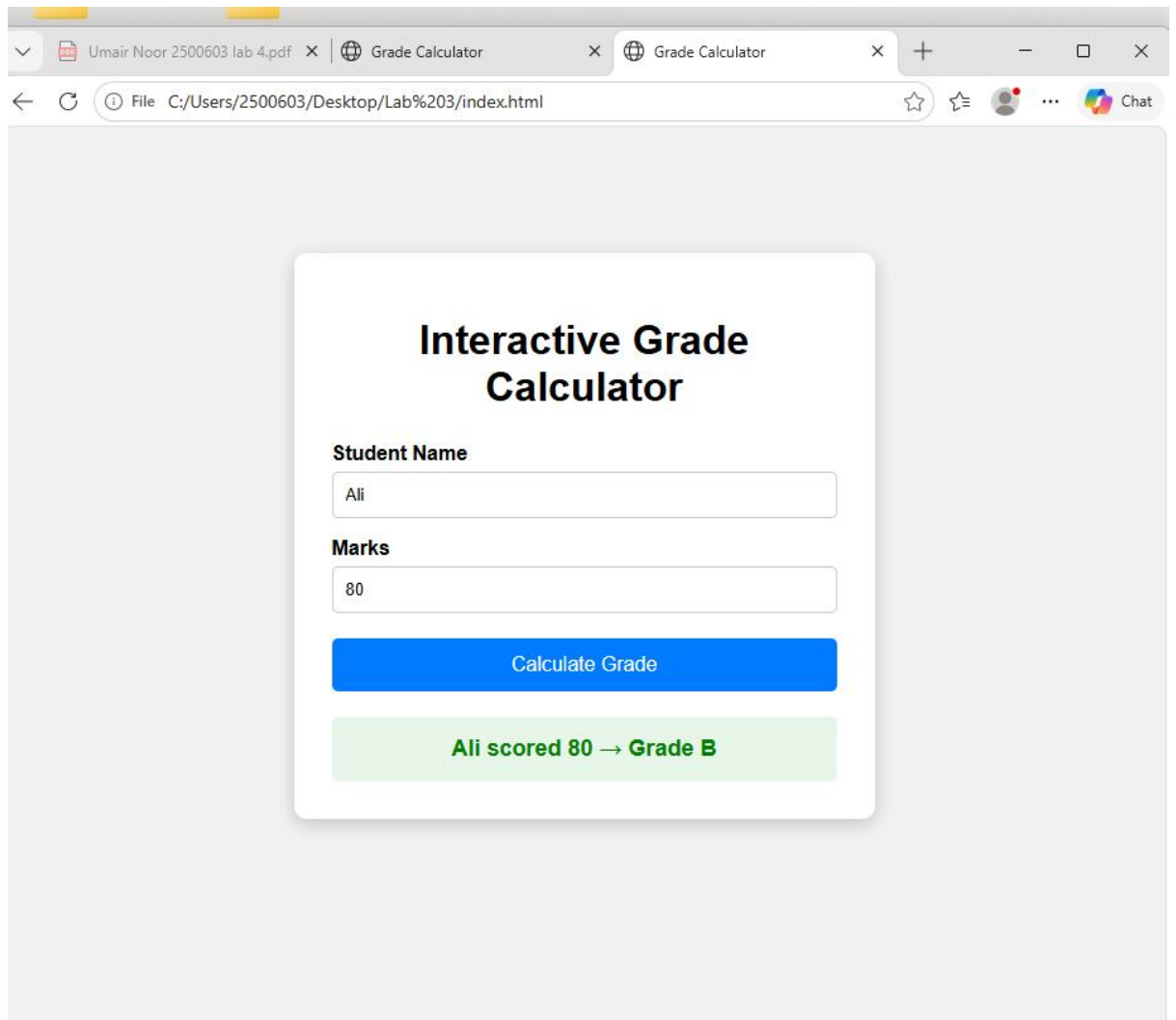
  // Assign grade using if / else if / else
  let grade;

  if (marks >= 85) {
    grade = "A";
  } else if (marks >= 70) {
    grade = "B";
  } else if (marks >= 50) {
    grade = "C";
  } else {
    grade = "F";
  }

  // Show result using template literal
  resultDiv.textContent = `${studentName} scored ${marks} = Grade ${grade}`;

  // Colour result green for pass and red for fail
  if (marks >= 50) {
    resultDiv.className = "pass";
  } else {
    resultDiv.className = "fail";
  }
});
});
```

Output Screenshots:



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File C:/Users/2500603/Desktop/Lab%203/index.html

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Chat

Interactive Grade Calculator

Student Name

Ahmed

Marks

60

Calculate Grade

Ahmed scored 60 → Grade C

Error Message :

Interactive Grade Calculator

Student Name

Ahmed

Marks

102

Calculate Grade

Marks must be between 0 and 100.